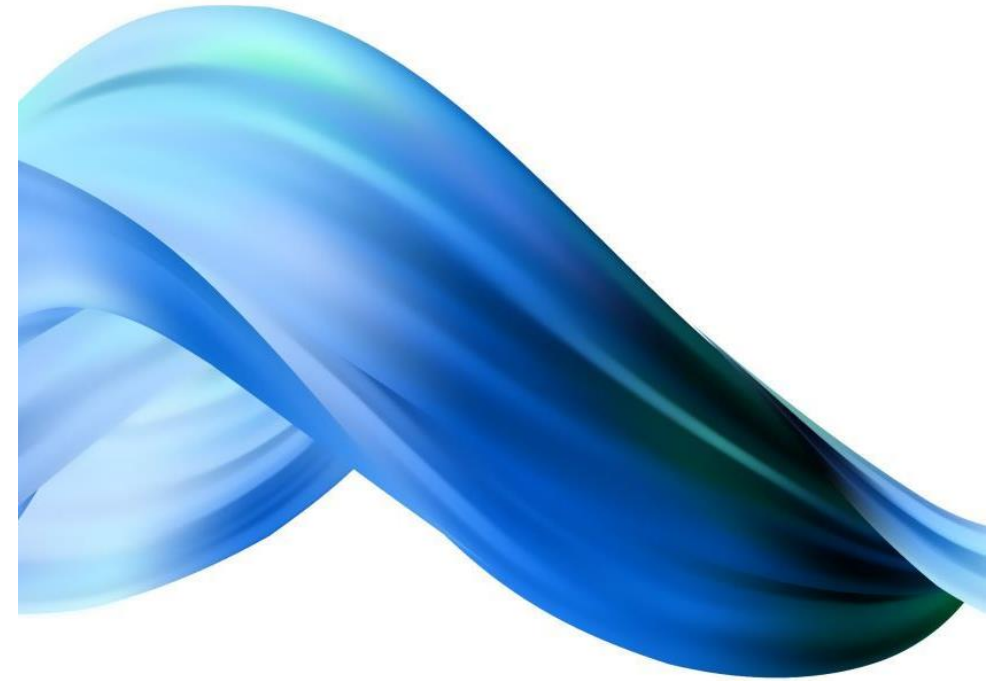

US HEALTHCARE PAYMENT INTEGRITY

CMS MEDICARE INPATIENT ANALYSIS

- Power BI:

<https://app.powerbi.com/view?r=eyJrIjoiMDY4YzV1NTQtOGFmZi00ZWYxLWIwMmMtZWQxYmE4YzUxZDk0IiwidCI6Ijg1ZDBjMTEyLTJjYzAtNDU1ZS04YjMxLTM1NDE1OWQzMjMDNkNiJ9>
- Python Pandas for data Cleaning
- Powerbi for Visualization
- By KIREETI AVULA



TASK 1 – DATA UNDERSTANDING



I have analyzed the datasets provided for the project. There are 3 tables in the data. The 1st table is **Inpatient Payments**, which contains hospital billing information such as hospital ID, procedure code, number of discharges, average charges submitted by the hospital, total payment amount and Medicare payment.



The second table is **DRG Details**, which gives information about the medical procedure like DRG code, diagnostic category and complexity of the treatment.

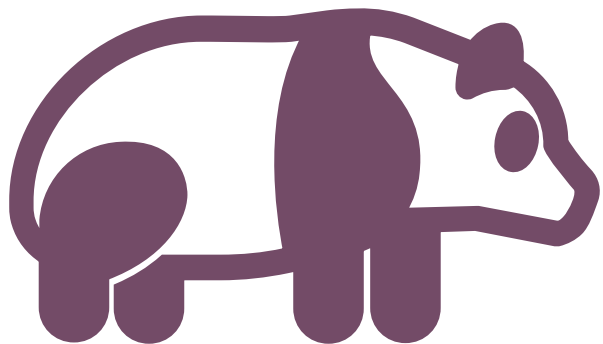


The third table is **Provider Info**, which contains hospital details like hospital name, ownership type, location and overall rating.



Using Python with **pandas** in **Google Colab**, I performed data cleaning and joined the tables using the common keys. After preparing the final dataset, I used **Power BI** to perform analysis and create visualizations

TASK 1 - DATA CLEANING



Data cleaning was done in Python using Pandas in Google Colab.

Steps performed:

- Loaded all three datasets into Pandas DataFrames
- Converted CCN (hospital IDs) to string format with leading zeros for correct joins
- Standardized ZIP codes and FIPS codes for geographic consistency
- Handled missing values in RUCA and MDC columns
- Removed empty and footnote columns
- Converted hospital rating column to numeric format
- Data Integration: Joined datasets using common keys.
- Billing Gap = Avg Submitted Charge – Avg Medicare Payment Charge Ratio = Avg Submitted Charge ÷ Avg Medicare Payment

Final Dataset: ~126K rows and 56 columns after cleaning and filtering. Exported for visualization and analysis.

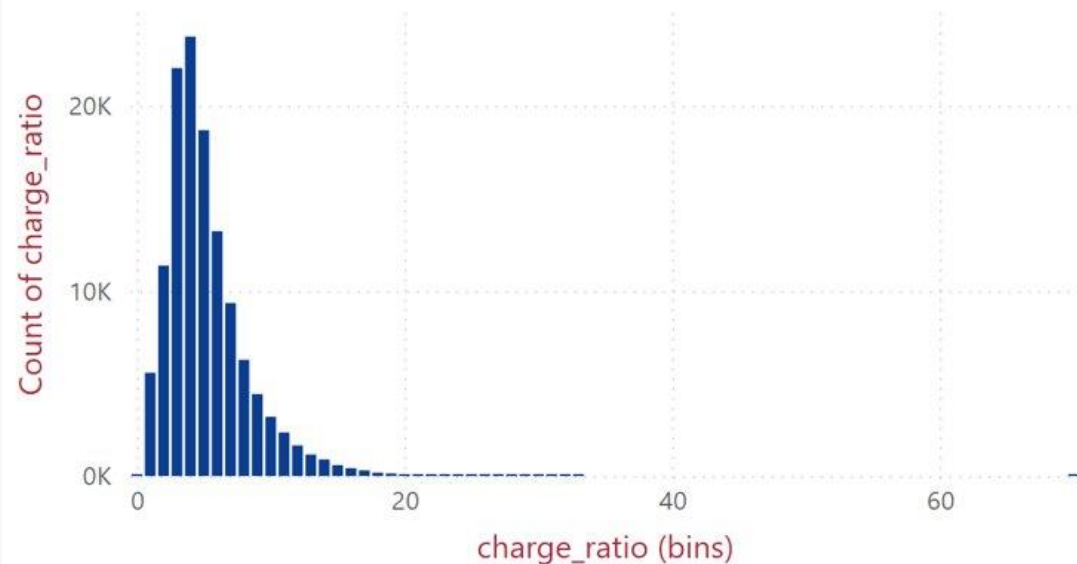
TASK 02 — THE BILLING GAP

- 2a — Top Hospitals by Billing Gap: Bar chart showing Top 20 hospitals with the highest billing gap. Stanford Health Care, Capital Health System and Cedars-Sinai are the top outliers — identified as more than 2 standard deviations above the national mean.
- 2b — Charge Ratio Distribution: Histogram showing national charge ratio distribution. Average charge ratio is 5.71 — hospitals bill nearly 6x more than Medicare pays. Distribution is strongly right-skewed with a long tail of extreme overbillers.
- 2c — Medicare Payments by State –MDC : Geographic map filtered by MDC showing how Medicare payments vary by state. Northeastern states and California show higher payment values due to higher case complexity and Medicare wage index adjustments.

Top 20 Hospitals by Billing Gap



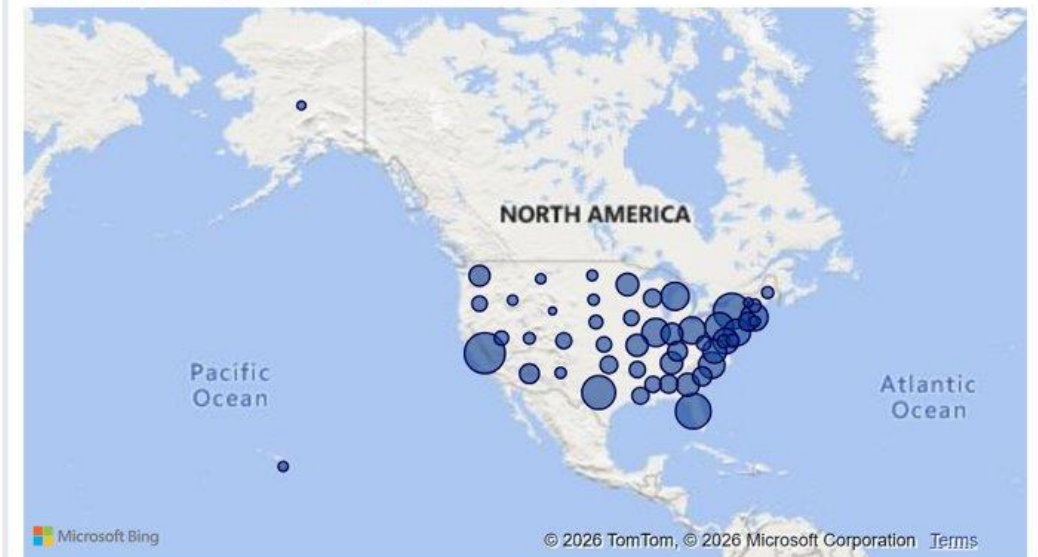
Charge Ratio Distribution



MDC

- ☐ 01
- ☐ 02
- ☐ 03
- ☐ 04
- ☐ 05
- ☐ 06
- ☐ 07
- ☐ 08
- ☐ 09

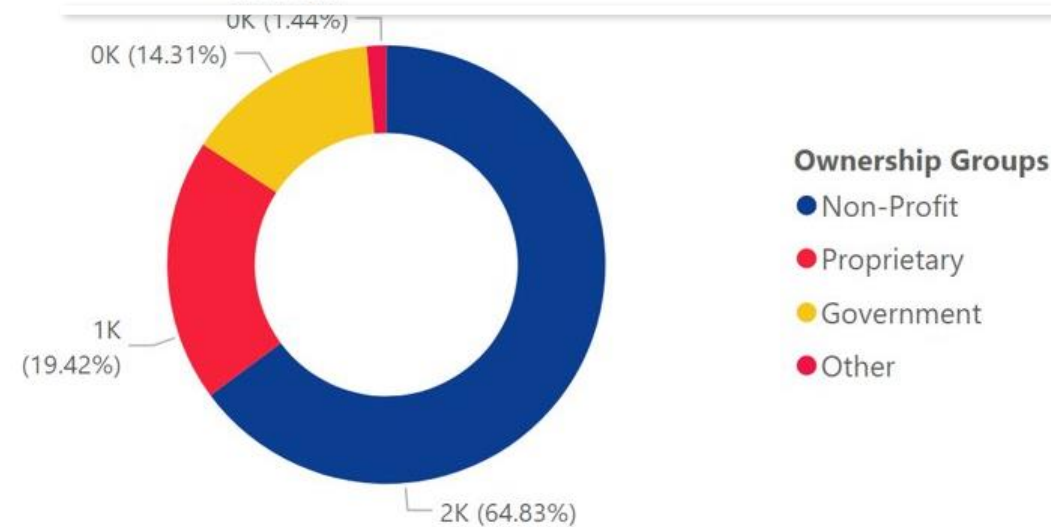
Medicare Payments By State - MDC



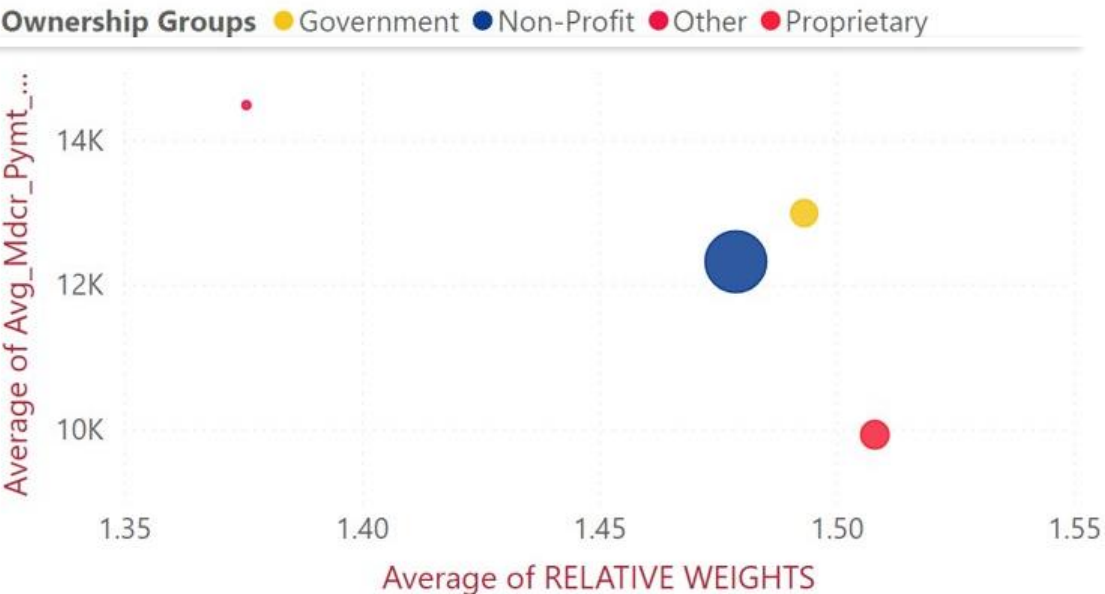
TASK 03 - OWNERSHIP AND SERVICES

- 3a — Ownership Mix: Donut chart showing 2,775 hospitals by ownership type. Voluntary non-profit hospitals dominate at nearly two-thirds of all facilities, followed by Government and Proprietary hospitals.
- 3b — MDC Mix by Ownership: Stacked bar chart of MDC distribution across ownership groups. Proprietary hospitals show a higher share of surgical MDCs — suggesting concentration on higher-revenue elective procedures. Dataset contains only Acute Care hospitals so ownership is the best segmentation.
- 3c — Relative Weight vs Medicare Payment: Scatter plot of DRG relative weight vs average Medicare payment. Non-profit hospitals show closer alignment between complexity and payment. Proprietary hospitals handle higher complexity but receive comparatively lower reimbursement.

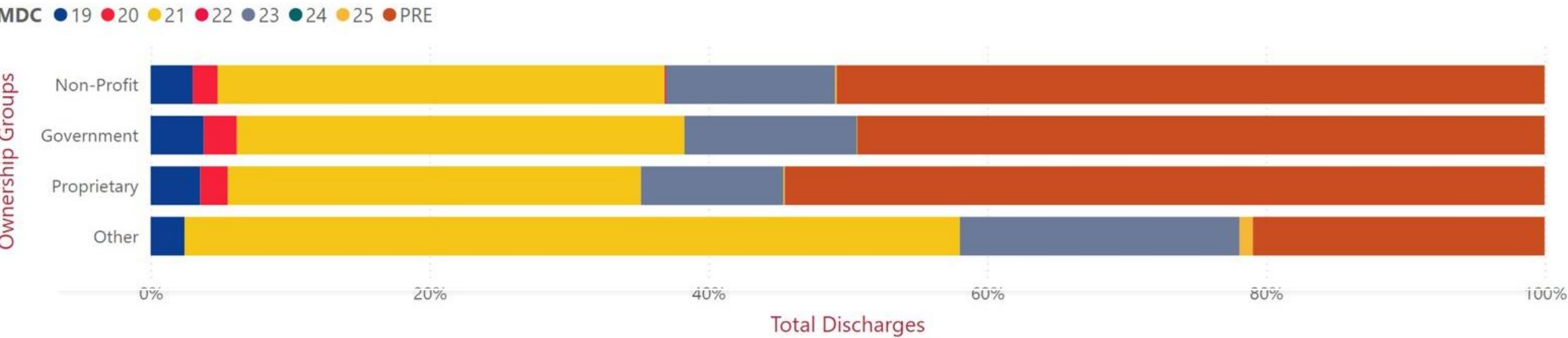
Hospital Ownership



Relative Wiegth vs Medicare Payment



MDC by Hospital Type



TASK 04 A - OWN FINDINGS

- Finding: For-Profit Hospitals Overbill Most for Surgery
- Proprietary (for-profit) hospitals have a materially higher average charge ratio compared to Non-Profit and Government hospitals.
- This gap widens for surgical DRG types versus medical cases — for-profit hospitals systematically overbill more for the same procedures.
- The decomposition tree further reveals the highest charge ratios are concentrated in specific states known for high Medicare fraud enforcement activity.

US Healthcare Payment Integrity

4M

Total Discharges

5.71

Avg Charge Ratio

51.56K

Avg Billing Gap

2775

Unique Hospitals

244.71bn

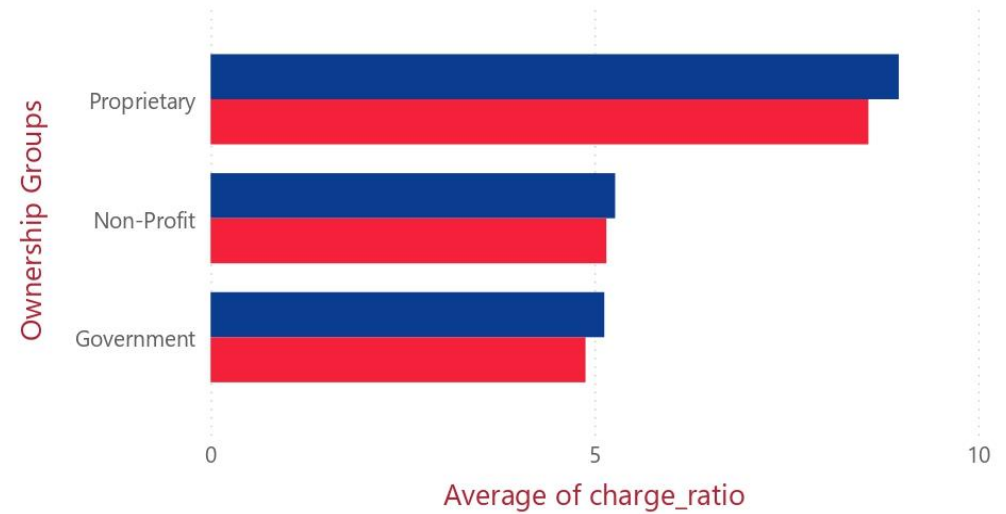
Total Billed

46.92bn

Total Medicare Paid

For-profit hospital overbill more for surgery

TYPE ● MED ● SURG



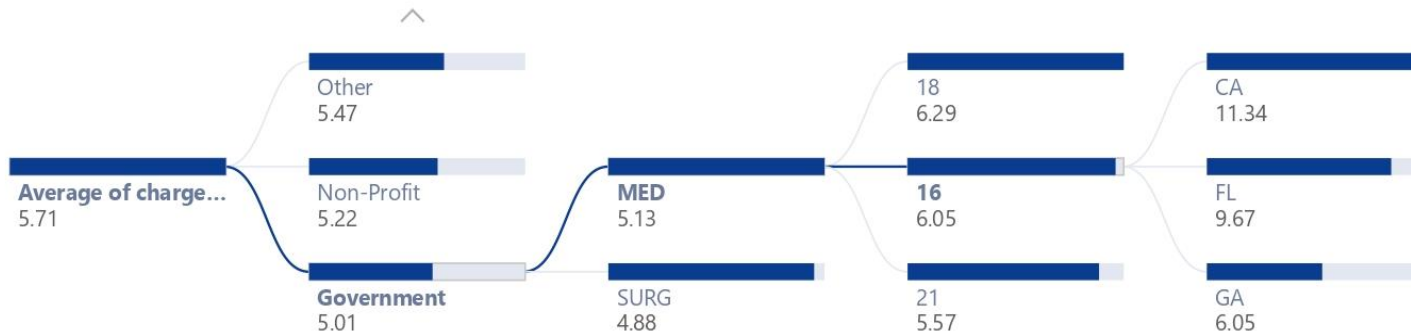
What Drives the highest charge ratio.?

Ownership Gr...
Government

TYPE
MED

MDC
16

Rndrng_Privr...



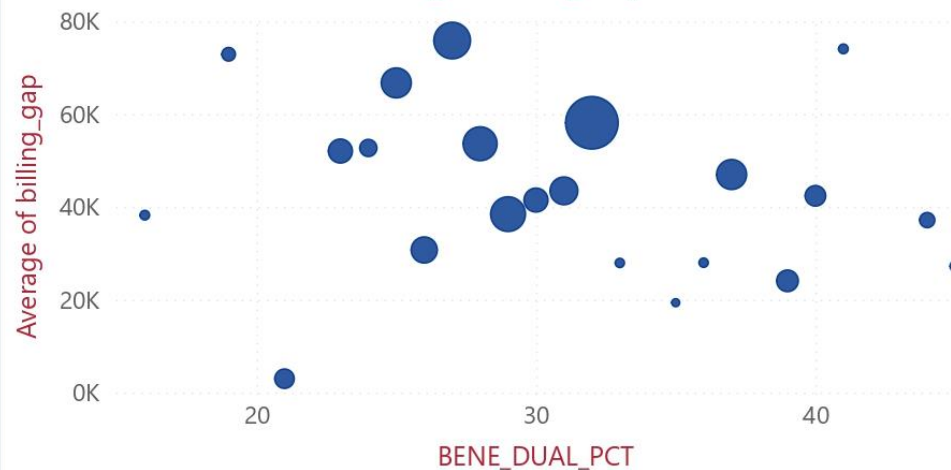
TASK 04 B - EXTERNAL DATABASE

- **External Data Source:** medicare-post-acute-care-hospice
- **Use File Source:** **data.cms.gov**
- Join Key: STATE → Rndrng_Privr_St
- Key Column: BENE_DUAL_PCT — percentage of patients enrolled in both Medicare and Medicaid (dual-eligible = poorest, most vulnerable beneficiaries)
- **States with the highest proportion of dual-eligible patients also show the highest average billing gaps.**
- <https://data.cms.gov/provider-summary-by-type-of-service/medicare-post-acute-care-hospice/medicare-post-acute-care-utilization-hospice/data/2017>

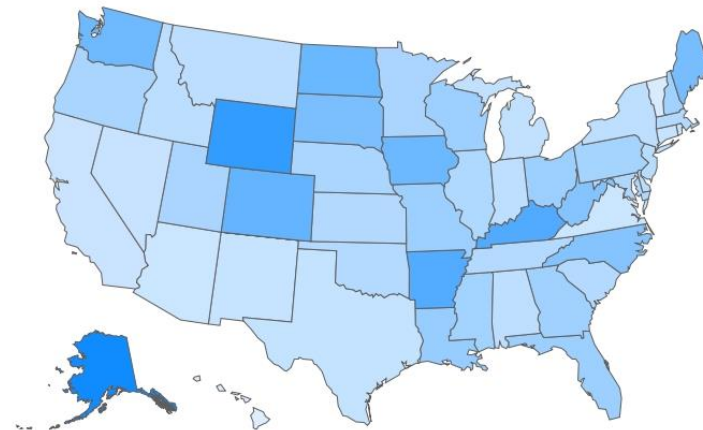
TASK 04B

- I joined CMS state-level data to check if vulnerable patients are linked to higher billing.
- BENE_DUAL_PCT captures dual-eligible patients — people covered by both Medicare and Medicaid, meaning the lowest income group.
- States with more dual-eligible patients show higher billing gaps. Hospitals are overbilling more in states where patients are financially weakest and least likely to question their bills.
- Overbilling is not random — it follows vulnerability.

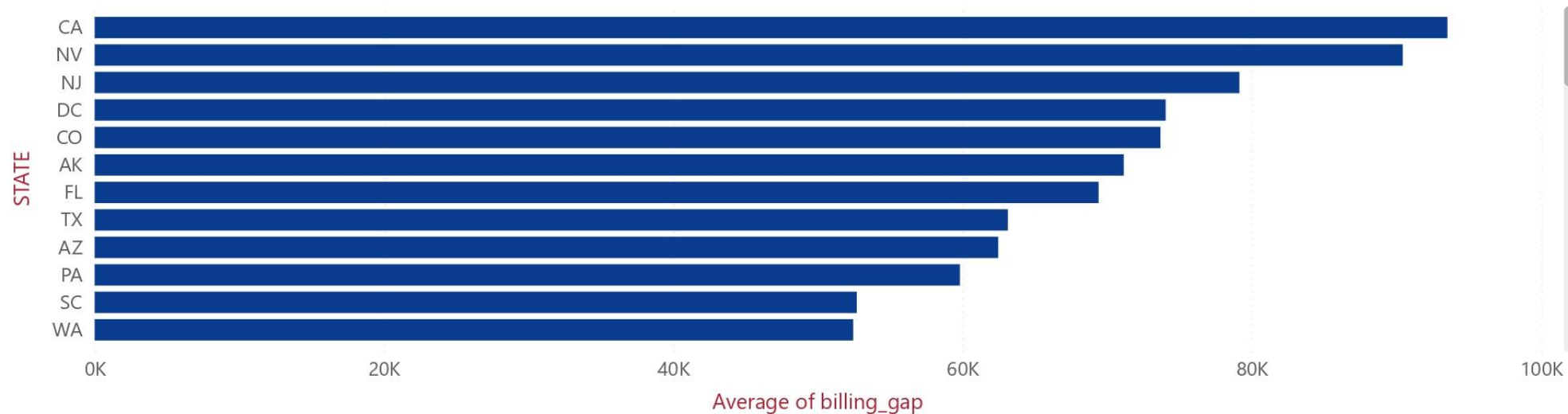
More Vulnerable Patients = Higher Billing Gap



Hospice Charge Ratio by State



Average of billing gap by STATE



THANK YOU

- Power BI Link

: <https://app.powerbi.com/view?r=eyJrIjoiMDY4YzVlNTQtOGFmZi00ZWYxLWIwMmMtZWQxYmE4YzUxZDk0IiwidCI6Ijg1ZDBjMTEyLTJjYzAtNDU1ZS04YjMxLTM1NDE1OWQzMjNkNiJ9>

